

Digital Forensics



Estimated time needed: 15 minutes

In this lab you will explore how you can investigate the logs to perform digital forensics on a system.

Learning Objectives

At the end of this lab you will be able to:

- Analyze the logs
- Use ssh to connect and analyze a remote system

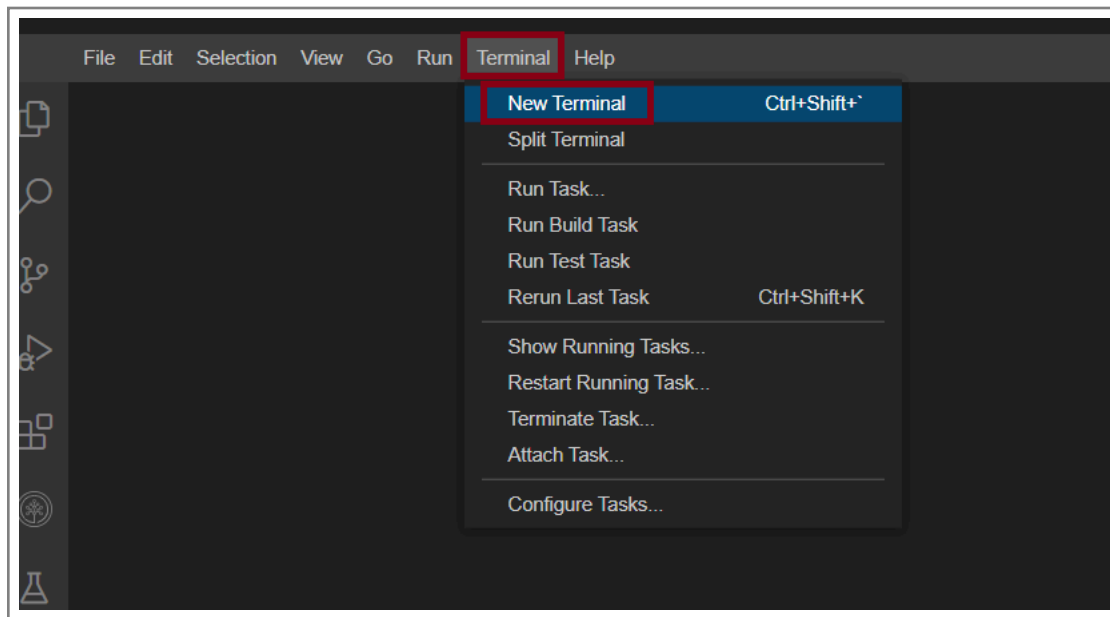
Prerequisites (optional)

It is good to be familiar with Linux environment. This lab requires Docker which is pre-installed in the lab environment.

Run Cowrie on docker

Cowrie is an SSH and Telnet honeypot which allows interactions designed to log brute force attacks and shell interaction, which in a real system is often performed by an attacker. This helps to analyze the time of attack, type of commands run by the attacker to perform the attack, other security issues and data breaches caused when the system was attacked, etc.,

1. Open a new terminal.



2. In the new terminal window, run the following command to run cowrie in a dockerized container.

```
docker run -p 2222:2222/tcp cowrie/cowrie > honeypotLogs.txt
```

This will start Cowrie and redirect all the logs which are printed on screen to the file named honeypotLogs.txt in your local machine.

Note: Please wait until the download finishes completely before moving to the next step.

Telnet to Cowrie

1. Open another terminal.
2. From the terminal, run the following command to telnet to Cowrie through ssh.

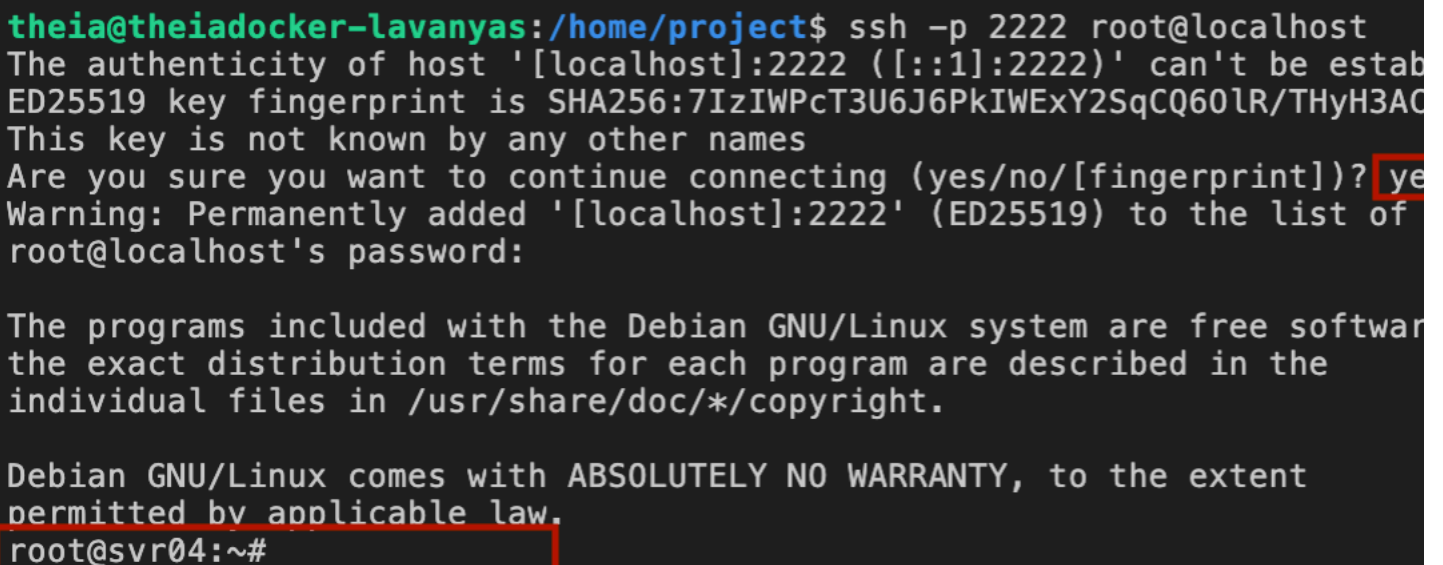
```
ssh -p 2222 root@localhost
```

3. When prompted to confirm Are you sure you want to continue connecting?, type **yes** and press enter.
4. When prompted for root@localhost password, type **admin** and press enter.

Note: When entering a password in the terminal, no characters or symbols will appear on the screen for security reasons. This is normal behavior — type the password and press Enter.

The connection will now be established and you will see the root prompt.

This session times out in 3-4 minutes if not used.



```
theia@theiadocker-lavanyas:/home/project$ ssh -p 2222 root@localhost
The authenticity of host '[localhost]:2222 ([::1]:2222)' can't be estab
ED25519 key fingerprint is SHA256:7IzIWPcT3U6J6PkIWExY2SqCQ60lR/THyH3AC
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? ye
Warning: Permanently added '[localhost]:2222' (ED25519) to the list of
root@localhost's password:

The programs included with the Debian GNU/Linux system are free softwar
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@svr04:~#
```

5. Now you will be connected to Cowrie and can run commands that can be executed on Cowrie. Run the following command in the ssh terminal.

```
ls
```

6. Create a file in the remote system by redirecting echo into a file.

```
echo "The is a file created through ssh">newfilefromRemote.txt
```

7. Check if the file is created and its content.

```
cat newfilefromRemote.txt
```

You should be able to see the content of the file.

8. Delete the file by running the following command.

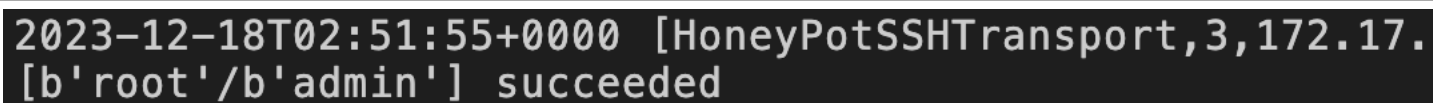
```
rm -f newfilefromRemote.txt
```

Analyze logs

1. Open a new Terminal.
2. In the terminal, run the following commands to check when the ssh request was received.

```
cat honeypotLogs.txt | grep "login attempt"
```

You will see the details about one or more login attempts to Cowrie as shown in the image below.

A terminal window with a black background and white text. The text shows a timestamp and a log entry for a successful SSH login attempt.

```
2023-12-18T02:51:55+0000 [HoneyPotSSHTransport,3,172.17.  
[b'root'/b'admin'] succeeded
```

3. Run the following command to check the logs for all the commands executed on the remote terminal.

```
cat honeypotLogs.txt | grep "CMD"
```

This will list all the commands executed in the ssh shell as given below.

```
theia@theiadocker-lavanyas:/home/project$ cat honeypotLo
2023-12-18T02:52:24+0000 [HoneyPotSSHTransport,3,172.17.
new file being created from the remote computer">newFile
2023-12-18T02:52:33+0000 [HoneyPotSSHTransport,3,172.17.
Remote.txt
2023-12-18T02:52:42+0000 [HoneyPotSSHTransport,3,172.17.
emote.txt
2023-12-18T02:52:47+0000 [HoneyPotSSHTransport,3,172.17.
mote.txt
2023-12-18T02:52:49+0000 [HoneyPotSSHTransport,3,172.17.
theia@theiadocker-lavanyas:/home/project$
```

4. Wait for 3 to 4 mins and execute the following command to see how the remote host closes the connection automatically.

```
cat honeypotLogs.txt | grep "Connection lost after"
```

```
2023-12-18T01:38:49+0000 [HoneyPotSSHTransport,0,172.17.
Connection lost after 40 seconds
2023-12-18T01:42:05+0000 [HoneyPotSSHTransport,1,172.17.
Connection lost after 189 seconds
2023-12-18T02:23:40+0000 [HoneyPotSSHTransport,2,172.17.
Connection lost after 184 seconds
2023-12-18T02:54:56+0000 [HoneyPotSSHTransport,3,172.17.
Connection lost after 182 seconds
```

Terminate Cowrie

1. Go to the original shell where you created the docker instance of Cowrie.
2. Press `ctrl + c` to terminate the instance.

Practice Exercises:

1. Run another Cowrie instance with docker.

```
docker run -p 2223:2222/tcp cowrie/cowrie >> honeypotLogs.txt
```

Note: The >> ensures that the logs are appended and not over written.

3. Telnet from 2 different terminal to the instance using same credentials **admin**.
4. Run some commands in both. Analyze the logs in honeypotlogs.txt.
5. Run the following command in a new terminal to see how long it takes for the telnet connections to be terminated.

```
tail -f honeypotLogs.txt
```

Here you are simulating analyzing logs of multiple telnet connections.

Conclusion

Congratulations! In this lab, you learned to telnet into a Cowrie instance running on a dockerized container and analyze the logs. In real life, you have various tools that make this log analysis process more detailed yet simpler.

Author(s)

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